Polynomial Regression Assignment

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1 Introduction

This assignment applies polynomial regression to two prediction problems. Var1 predicts Net Power Score from six input variables, while Var2 predicts Thermal Anomaly Score from three spatial variables.

Polynomial degree and Elastic Net parameters are selected using five-fold cross-validation. The selected models are then trained on the complete training datasets and used to predict the hidden test sets.

The complete implementation, including the training and prediction code, saved models, plots, and prediction files, is available in the accompanying **GitHub repository**.

2 Methodology

2.1 Polynomial Regression

Polynomial features and interaction terms are generated using `PolynomialFeatures`. Degrees 1–10 are tested for Var1 and degrees 1–20 for Var2.

The generated features are standardized using `StandardScaler`. No min-max normalization is applied because the original inputs are already on a comparable scale.

2.2 Elastic Net and Model Selection

Elastic Net combines L1 and L2 regularization. The regularization strength is controlled by α, while β denotes the L1 ratio.

$$\text{Loss} + \alpha \left[\beta \|w\|_1 + (1-\beta)\|w\|_2^2\right].$$

Five-fold shuffled cross-validation with `random_state=42` is used to evaluate candidate degrees and Elastic Net hyperparameters. For each degree, the best validation MSE is recorded. The configuration with the lowest CV MSE is selected and the final model is retrained using all 1,000 training observations.

2.3 Evaluation

The primary evaluation metric is Mean Squared Error:

$$\text{MSE} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2.$$

Training R^2 is also recorded. Since the test targets are hidden, test-set MSE and R^2 cannot be calculated locally.

3 Results

3.1 Var1 – Net Power Score

The selected Var1 model uses polynomial degree 5.

Table 1: Var1 selected model

Degree	5
α	0.0083268
β	0.95
CV MSE	0.333396
Training MSE	0.227370
Training R^2	0.976938

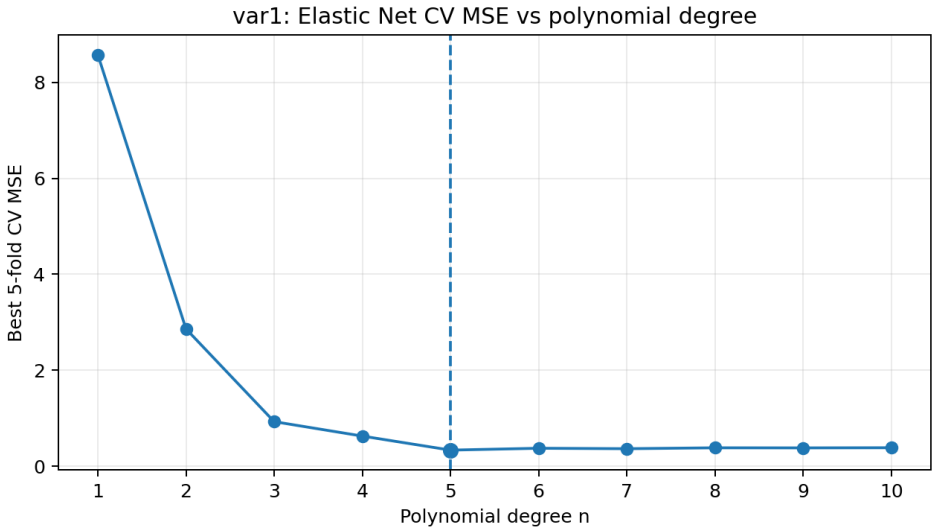


Figure 1: Var1 CV MSE versus polynomial degree.

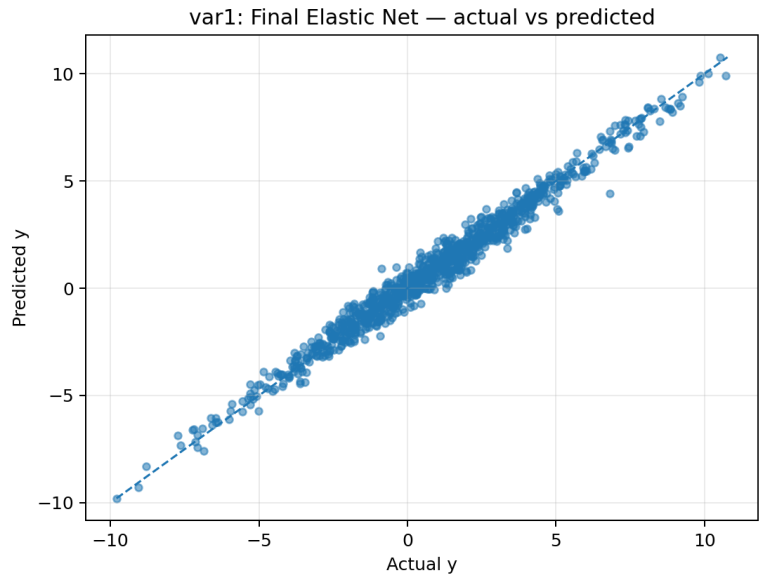


Figure 2: Var1 actual versus predicted values.

3.2 Var2 – Thermal Anomaly Score

The selected Var2 model uses polynomial degree 12.

Table 2: Var2 selected model

Degree	12
α	0.0057735
β	0.05
CV MSE	0.253945
Training MSE	0.189553
Training R^2	0.996291

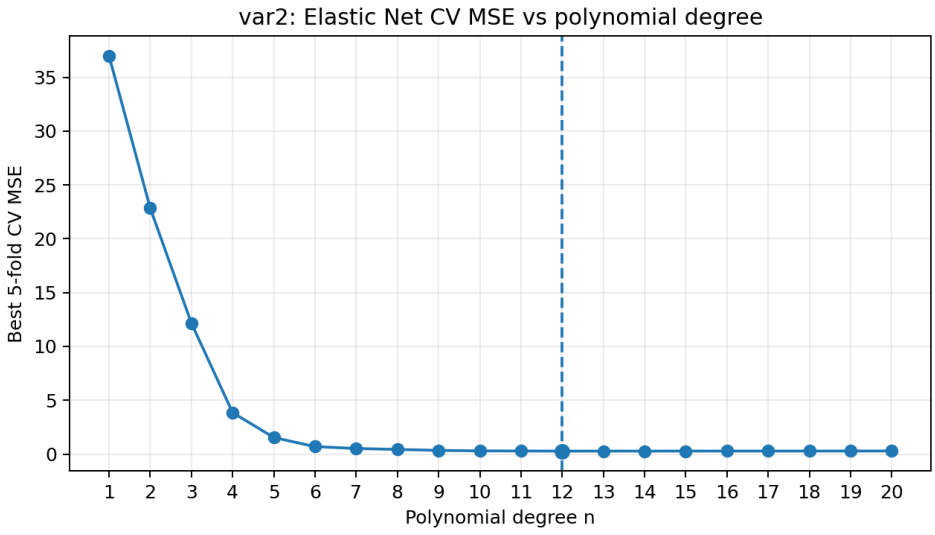


Figure 3: Var2 CV MSE versus polynomial degree.

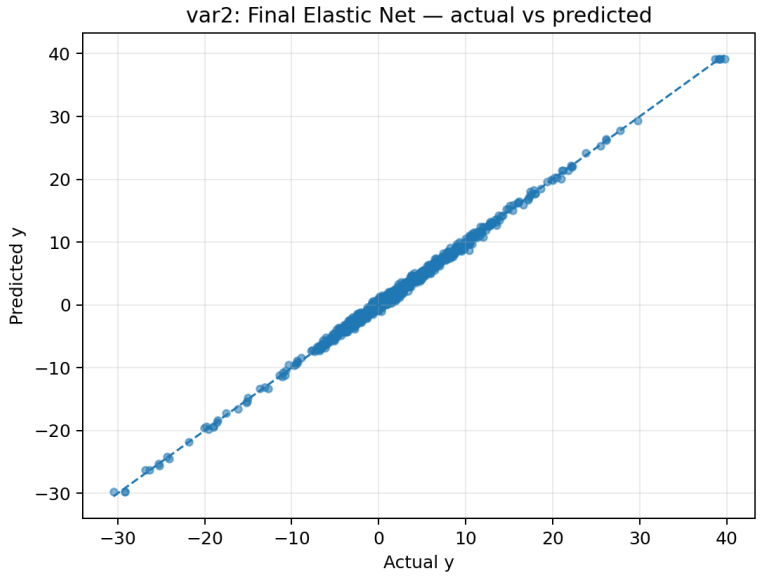


Figure 4: Var2 actual versus predicted values.

4 Final Model Selection

Table 3: Final selected models

Problem	Degree	α	β	CV MSE
Var1	5	0.0083268	0.95	0.333396
Var2	12	0.0057735	0.05	0.253945

The final models were retrained using all 1,000 training observations and used to generate predictions for the hidden test sets.

5 Conclusion

Elastic Net polynomial regression was used for both prediction problems. Cross-validation selected degree 5 for Var1 and degree 12 for Var2, with CV MSE values of 0.333396 and 0.253945 respectively.

The selected models were trained on the complete training data and used to produce the required test predictions.